Innomatic Research lab

Assignment Tasks

Task 1: Conceptual Understanding

1. What is the difference between "Love" and "love" in NLP ?

In NLP "Love" (capital L) and "love" (small l) are treated as diff-rent tokens if case normalization is not applied.

- "Love" → often appears at the beginning of a sentence or a prop noun.
- "love" → common word expressing emotion

If we cannot convert everything to lowercase, the model thinks t are two separate words, think which :

- Increases vocabulary size unnecessarily
- reduces model efficiency

So most preprocessing pipelines convert all text to lowercase

2. What happens if stopwords are not removed ?

If stopwards (like is, the and, of) are not removed :

- Noise increases → these words don't add much meaning
- Model becomes slower → more words to process
- Accuracy may drop → important words gets diluted
- storage increases → larger vocabulary

But don't blindly remove them — sometimes they matter

3. Two real-world scenarios where removing stopwords is harmful

1. Sentiment analysis
   - Example : "I do not like this product"
   - If you remove "not", it becomes "I like this product" → completely wrong meaning

2. Question Answering / Search Queries
   - Example : "What is the best phone?"
   - Removing stopwords may distort structure and intent
   - Words like "what", "is" help understand the question type

So removing stopwords blindly = ~~bt~~ bad preprocessing

4. Difference between stemming and lemmatization

Stemming
- Cuts words to root from using simple rules
- often produces incorrect or non-real words
- faster, but less accurate
- Example : "running" → "run", "studies" → "studi"

Lemmatization
- Convert words to their actual dictionary base form (lemma)
- Uses grammar and vocabulary knowledge
- slower, but more accurate
- Example : "running" → "run", "better" → "good"

In short :
- stemming = quick but messy
- lemmatization = slower but correct